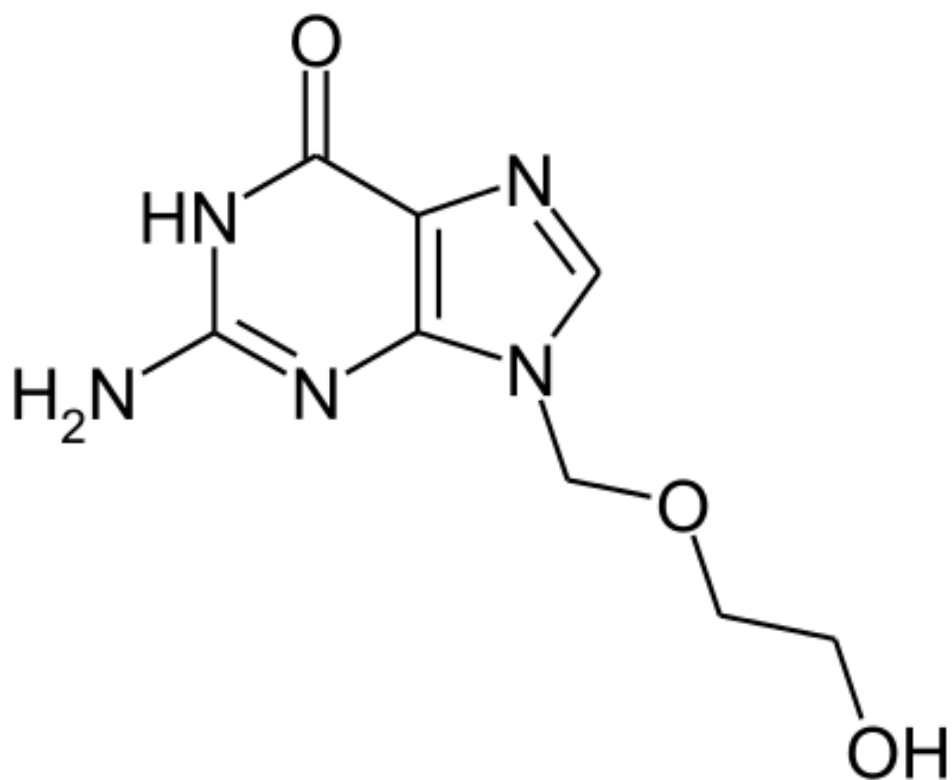




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ACICLOVIR DYNAMICS:  
A MATHEMATICAL MODEL

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# Summary

In this report, the development process of a model capable of describing the dynamics of the antiviral drug aciclovir throughout the gastrointestinal tract and the blood is presented. The model is composed of differential equations and the simulations of such model were performed using Euler's method. The model proposed gives accurate predictions that comply with experimental data.

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# List of variables

$t$ : time.

$M_{GI}(t)$ : Aciclovir mass present in the contents of the gastrointestinal tract at time  $t$ .

$M_{blood}(t)$ : Aciclovir mass present in the blood at time  $t$ .

$V_{GI}(t)$ : Volume of the contents of the gastrointestinal tract at time  $t$ .

# Chapter 1

## Introduction

Aciclovir (or acyclovir) is an antiviral drug used to combat the members of the herpesvirus family. The intention of this report is to present the development process of a model capable of describing the dynamics of aciclovir throughout the body; namely, how aciclovir taken orally in the form of pills is diffused from the gastrointestinal tract to the bloodstream, and how this drug is then eliminated from the body. This model could aid doctors in prescribing the correct dosage of aciclovir for a given patient. The model takes into consideration several important factors. The first one is the bioavailability; the percentage of medicine which trespasses the gastrointestinal tract barrier and reaches the bloodstream. The inclusion of bioavailability into the model is specially important for the case of aciclovir since it is reported to lie between the low values of 10% and 20%. [1] The second one is a variable volume of gastrointestinal tract contents; which is mostly altered by ingestion. The third and last one is the dissolution of the pills in the stomach. Chapter 2 deals with the first modeling cycle, in which the change in the mass of aciclovir in the blood with respect to time was modeled by a single differential equation assuming the concentration of aciclovir in the contents of the gastrointestinal tract to be constant. In chapter 3 this concentration was allowed to vary with time. In chapter 4 bioavailability was incorporated to the model. In chapter 5 the volume of the gastrointestinal tract contents was allowed to vary with time. Finally, in chapter 6 the effects of the dissolution of pills in the stomach was considered.

## Chapter 2

# Dynamics of aciclovir considering a constant concentration within gastrointestinal tract contents

The purpose of this chapter is to present a first model that describes how aciclovir is transported from the gastrointestinal tract to the blood and then eliminated by making a gross assumption: that the concentration of aciclovir within the gastrointestinal tract contents is constant. This first model is very far from accurately describing the situation presented in the introduction. The model does not take into account the consumption of discrete quantities of medicine in the form of pills, but rather a continuous intake such that the concentration of aciclovir in the contents of the gastrointestinal tract remains constant. Section 2.1 introduces the differential equations that model the situation and explains the meaning of each variable and constant in it. Section 2.2 deals with the selection of appropriate values for the constants in the model. Finally, in section 2.3, the results of the simulation using Euler's method are presented.

### 2.1 The differential equation

$$\frac{dM_{blood}(t)}{dt} = d_1 \left( C_{GI} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \quad (2.1)$$

$$C_{GI} = \frac{M_{GI}}{V_{GI}}$$



Where  $t$  is time,  $M_{blood}(t)$  is the aciclovir mass present in the blood at time  $t$ ,  $C_{GI}$  is the concentration of aciclovir in the contents of the gastrointestinal tract,  $V_{blood}$  is the blood volume in the body,  $d_1$  is the transfer rate of the drug from the gastrointestinal tract to the blood,  $d_2$  is the clearance of the drug from the blood,  $M_{GI}$  is the aciclovir mass present in the gastrointestinal tract and  $V_{GI}$  is the volume of gastrointestinal tract contents.

## 2.2 Selection and determination of appropriate values for the constants

Constants were selected to fit an average adult in a fasting condition. If this model were to be applied to a specific patient, the values of the constants would have to be changed; it is evident that some constants are far more variable from patient to patient while other ones are much more conserved among them.

As pills for treating Herpes Zoster (commonly known as Shingles) generally contain 800 mg of aciclovir, this mass was considered to be present in the gastrointestinal tract ( $M_{GI}$ ). As for the volume of the gastrointestinal tract contents ( $V_{GI}$ ), it was reported to be, in average, 0.163 L with a standard deviation of 0.102 L while fasting. [2] An average man has about between 4.5 and 5.5 L of blood (averaging to about 5 L) running through his body ( $V_{blood}$ ). [3]

The transfer rate of aciclovir from the gastrointestinal tract to the blood ( $d_1$ ) was not found in literature. This first model does not resemble the real behavior of the phenomena, because of this, the interest was not to obtain values that compare to experimental results. Then, for this first cycle the constant was trivially defined to be 1 L/h. For now, the clearance for aciclovir ( $d_2$ ) was also trivially defined to be 0.3 L/h.

## 2.3 Simulation

Euler's method was implemented using Python and the numpy library. A simulation of equation 2.1 with  $M_{blood}(0) = 0$  was then run.

The most important thing to realize from figure 2.1 is that  $M_{blood}(t)$  reaches an stable equilibrium.

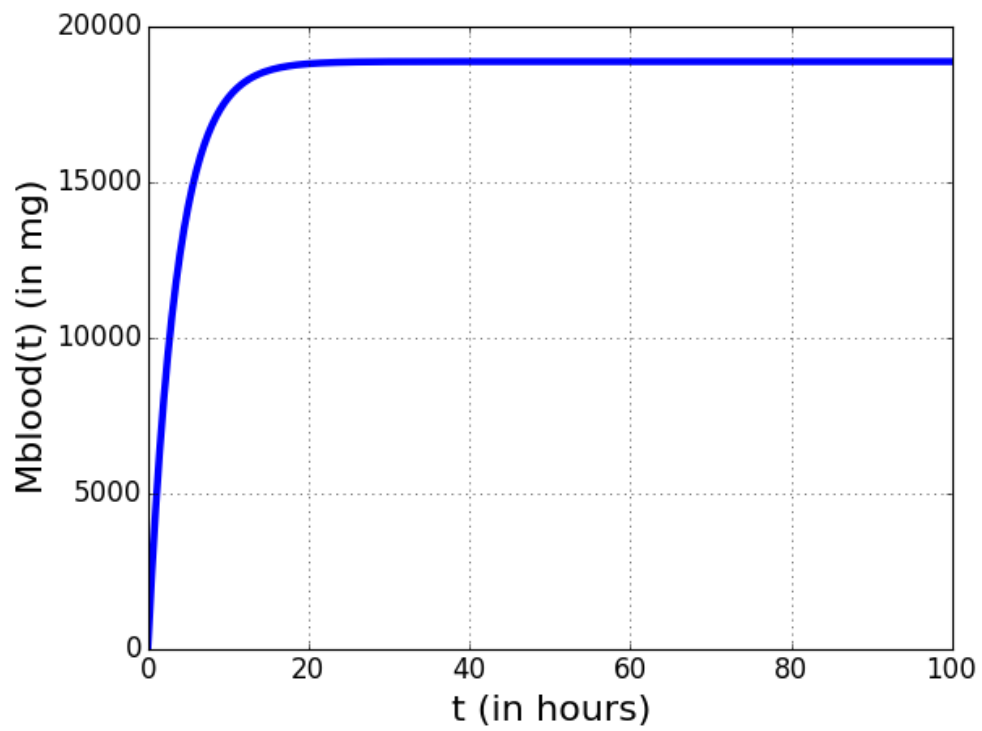


Figure 2.1: **Variation of aciclovir mass in blood with respect to time according to eq. 2.1** The values used for the constants where the ones presented in section 2.1

## Chapter 3

# Dynamics of aciclovir considering a variable concentration within gastrointestinal tract contents

The purpose of this chapter is to present a second model; one that describes how aciclovir is transported from the gastrointestinal tract to the blood and then eliminated considering a variable concentration of aciclovir in the contents of the gastrointestinal tract. Because this second model still does not take into account the bioavailability, the results will not be accurate, but the behavior will be close to that of the real phenomena. Section 3.1 introduces the system of differential equations that model the situation. In section 3.2 an appropriate value for the transfer rate of the drug from the gastrointestinal tract to the blood ( $d_1$ ) is determined to fit experimental data. In section 3.3, the results of several simulations using Euler's method are presented.

### 3.1 The system of differential equations

$$\frac{dM_{GI}(t)}{dt} = -d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \quad (3.1)$$

$$\frac{dM_{blood}(t)}{dt} = d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \quad (3.2)$$

Where  $M_{GI}(t)$  is the aciclovir mass present in the gastrointestinal tract at time  $t$  and  $V_{GI}$  is the volume of the contents of the gastrointestinal tract.

### 3.2 Determination of the transfer rate of aciclovir from the gastrointestinal tract to the blood

The data recovered by P. Susantakumar, *et al.* for patients who were given 800 mg of aciclovir in fasting conditions, shows that the mean peak aciclovir concentration of the drug in the blood was reached at about 2 h after the dosage. [4] When a value of 0.2 L/h for the transfer rate of aciclovir from the gastrointestinal tract to the blood ( $d_1$ ) was used, while maintaining the clearance ( $d_1$ ) at the value of 0.3 L/h, the model indeed reached the peak aciclovir concentration at about two hours after the dosage.

### 3.3 Simulations

A simulation of the system composed of equations 3.1 and 3.2 with  $M_{GI}(0) = 800mg$  and  $M_{blood}(0) = 0$  was run; the result is shown in figure 3.1.

It is usually indicated for adults who suffer a shingles episode to take pills that contain 800 mg of aciclovir every 4 h for 7 to 10 days. [5] Figure 3.2 shows the first 24 h of a treatment. Figure 3.3 shows a complete treatment consisting of 7 days; the simulation was ran until the tenth day. As time passed, the aciclovir mass in the blood started oscillating with respect to a fixed point of approximately 3219.5 mg with a deviation of around  $\pm 195.5$ . When the dosage was cut at the seventh day, the process of completely clearing the aciclovir from the blood was started. After a while, both the aciclovir mass in the blood and in the contents of the gastrointestinal tract started tending more and more to zero. Therefore 0 is a stable equilibrium point for the aciclovir mass in the blood and in the contents of the gastrointestinal tract.

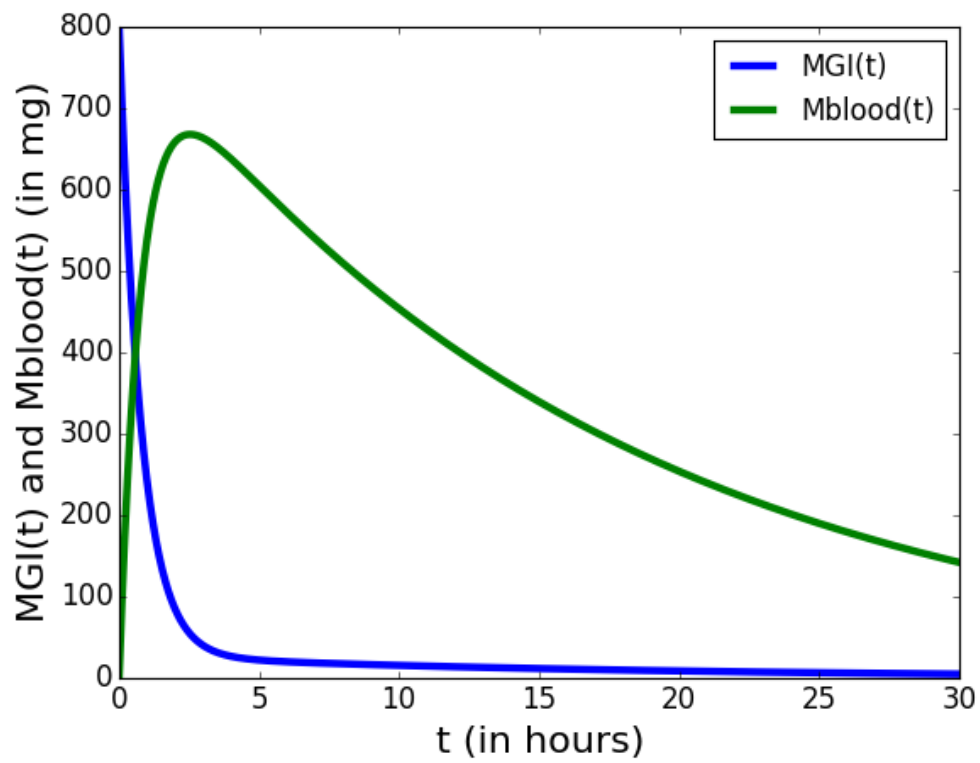


Figure 3.1: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 3.1 and 3.2 Single dosage

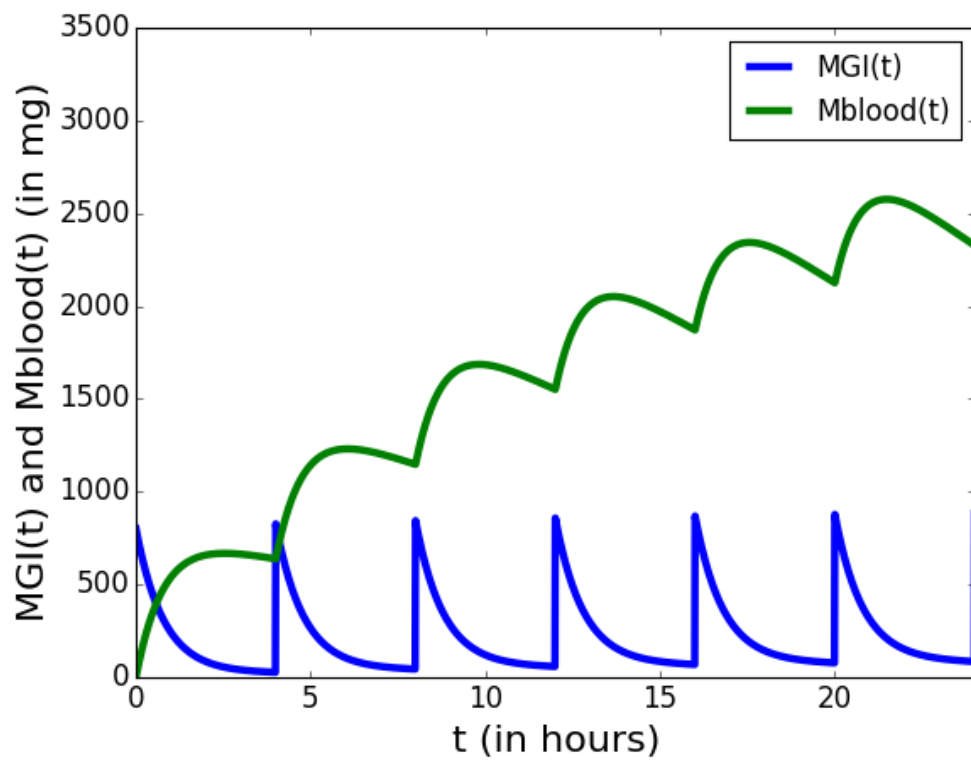


Figure 3.2: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 3.1 and 3.2 First 24 h of a usual 7 day treatment

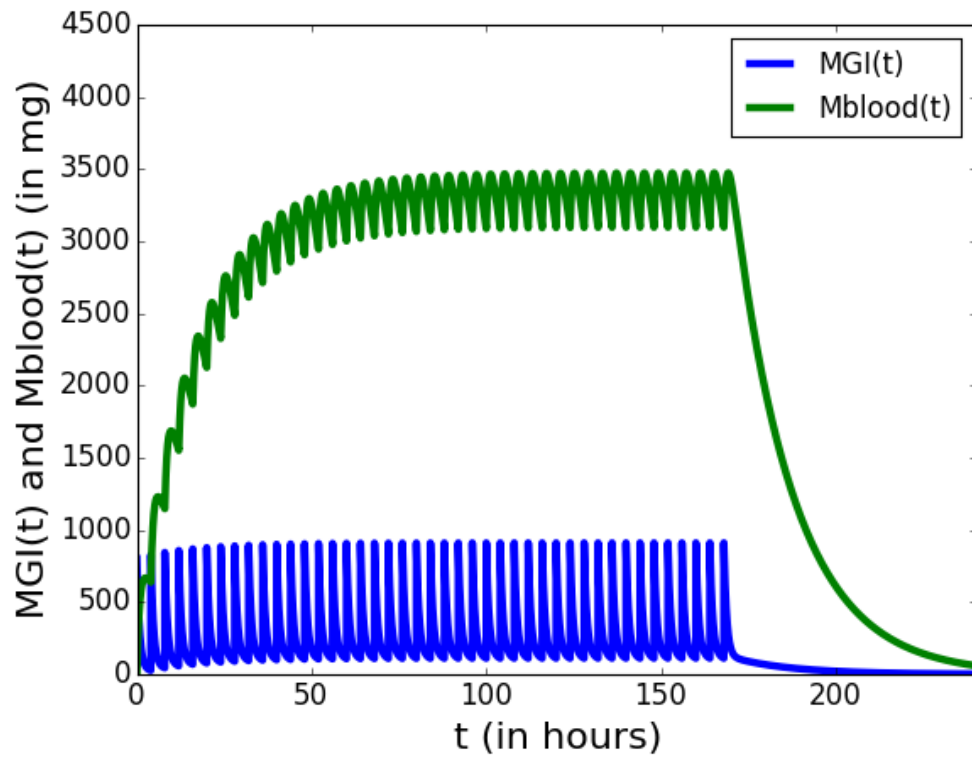


Figure 3.3: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 3.1 and 3.2 Complete usual 7 day treatment and the three of following days

## Chapter 4

# Incorporation of bioavailability

In this chapter the model is modified to account for bioavailability; which is necessary for the model to yield results comparable to those obtained experimentally. Section 4.1 introduces the new system of differential equations. In section 4.2 the modification made to the model is validated. Section 4.3 deals with the selection of values for the bioavailability (a new constant;  $d_3$ ) and the clearance ( $d_2$ ), as well as the determination of an appropriate transfer rate of aciclovir from the gastrointestinal tract to the blood ( $d_2$ ) to fit experimental data. Lastly, in section 4.4 the results of several simulations are presented.

### 4.1 The system of differential equations

The bioavailability is the percentage of medicine which trespasses the gastrointestinal tract barrier and reaches the bloodstream. For non-intravenous medicine this percentage is less than a 100%. Thus, a modification to the systems of differential equations presented in chapter 3 has to be made. The bioavailability ( $d_3$ ) has to multiply the first term of equation 3.2; yielding equation 4.2. Note that equations 3.1 is kept unchanged so it is the same as equation 4.1.

$$\frac{dM_{GI}(t)}{dt} = -d_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \quad (4.1)$$

$$\frac{dM_{blood}(t)}{dt} = d_1 d_3 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \quad (4.2)$$



## 4.2 Modification validation

The bioavailability of aciclovir lies between 10% and 20%. [1] Generally, the value is about 20%. So if the bioavailability is set to 0.2 and we ignore the elimination of the drug from the blood (that is,  $d_2$  has a value of 0), the aciclovir in the blood should tend to a fifth of the initial concentration in the gastrointestinal tract as time goes to infinity. Because the initial quantity of aciclovir in the gastrointestinal tract was 800 mg, aciclovir in the blood should tend to 160 mg, as shown in figure 4.1.

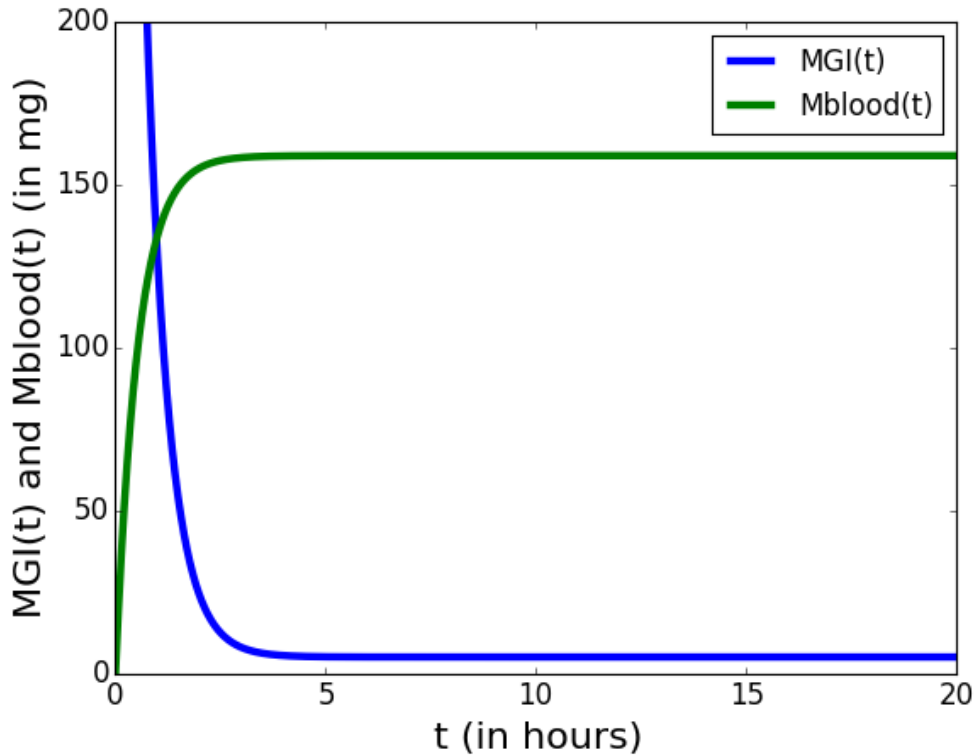


Figure 4.1: **Aciclovir mass in the blood with respect to time without considering drug elimination from the blood according to equations 4.1 and 4.2**  $d_1$  is equal to 0.1 L/h,  $d_2$  is equal to 0 and  $d_3$  is equal to 0.2

This means that when the elimination of the drug from the blood is taken into account, although the maximum amount of aciclovir in the blood at any given time would be less than 160 mg, those 160 mg should pass through the blood.

### 4.3 Selection and determination of appropriate values for the constants to fit experimental data

The clearance for aciclovir ( $d_2$ ) is reported to be 290 ml/min or 17.4 L/h for patients with a healthy kidney and a  $1.73 \text{ m}^2$  body surface area (which is average for 25 year old males). [6] The data recovered by P. Susantakumar, *et al.* for patients who were given 800 mg of aciclovir in fasting conditions, shows that the mean peak aciclovir concentration of the drug in the blood plasma was about 1100 ng/ml. [4] Blood is composed of about 55% plasma and 45% blood cells. [7] So an average person has about 2.75 L of plasma. Aciclovir then peaks at 3.025 mg in plasma, and assuming that aciclovir is distributed homogeneously throughout the whole blood, it peaks at 5.5 mg in blood. The value for the transfer rate of aciclovir from the gastrointestinal tract to the blood ( $d_2$ ) was selected so that it peaked at 5.5 mg. Such value is 0.0225 L/h.

### 4.4 Simulations

Figure 4.2 shows the simulation for the system composed of equations 4.1 and 4.2 for a single dose. To facilitate visualization,  $M_{GI}(t)$  was scaled in a 1:100 ratio. It was mentioned that according to the data recovered by P. Susantakumar, *et al.* the peak was reached at about two hours. In figure 4.2 the peak is reached at about one hour. This discrepancy may be due to the fact that the model in its current form does not account for the dissolution of the pill in the stomach. However, the model at this state is a major improvement when compared to that presented in the last chapter.

Figure 4.3 shows the simulation of the first 24 hours of the treatment described in section 3.3 using the enhanced model presented in this chapter. Figure 4.4 shows the same simulation but till the tenth day. This time, for figures 4.3 and 4.4,  $M_{GI}(t)$  was scaled in a 1:50 ratio.

The initial conditions were again set to be  $M_{GI}(0) = 800\text{mg}$  and  $M_{blood}(0) = 0$ .

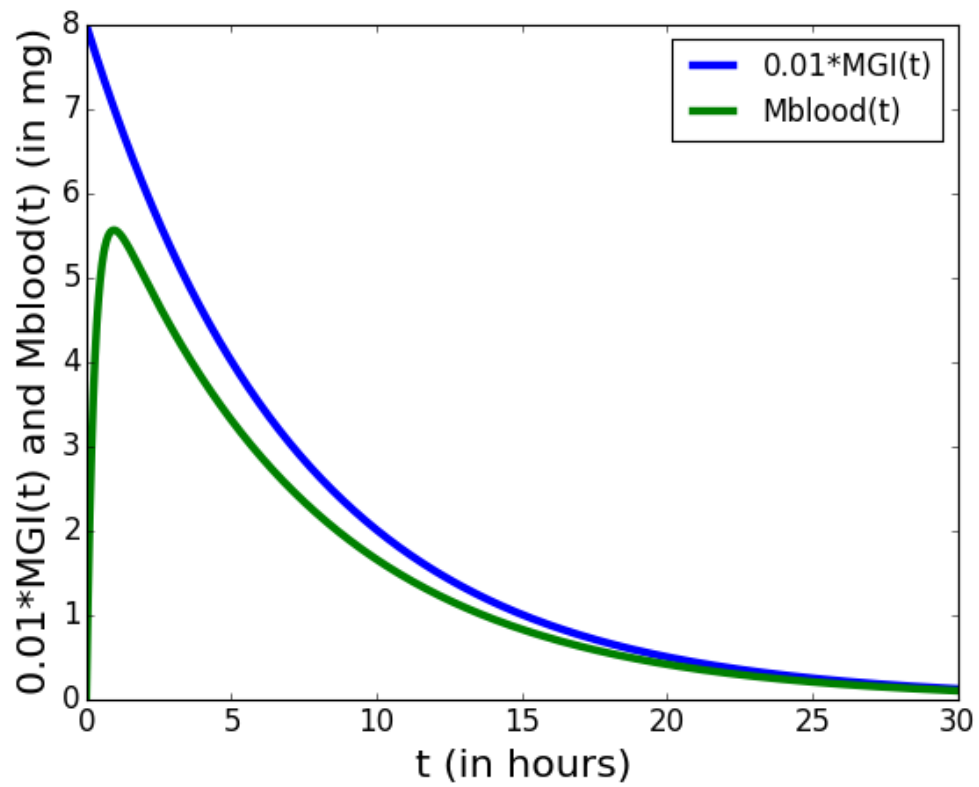


Figure 4.2: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 4.1 and 4.2 Single dosage;  $M_{GI}(t)$  was scaled in a 1:100 ratio

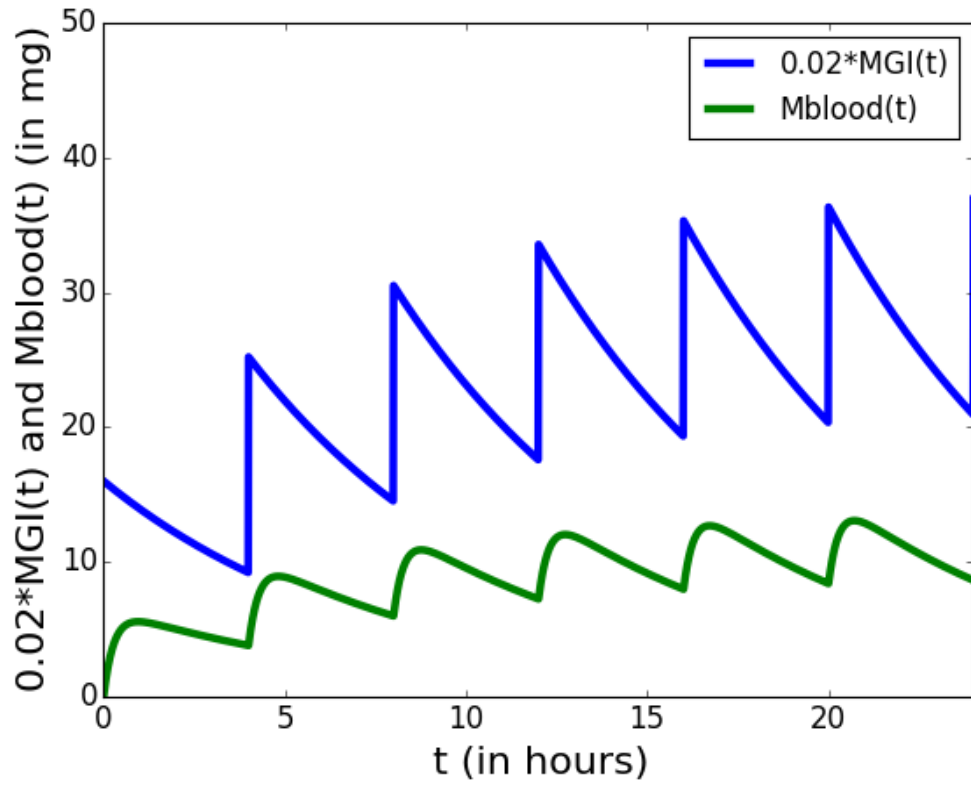


Figure 4.3: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 4.1 and 4.2 First 24 h of a usual 7 day treatment;  $M_{GI}(t)$  was scaled in a 1:50 ratio

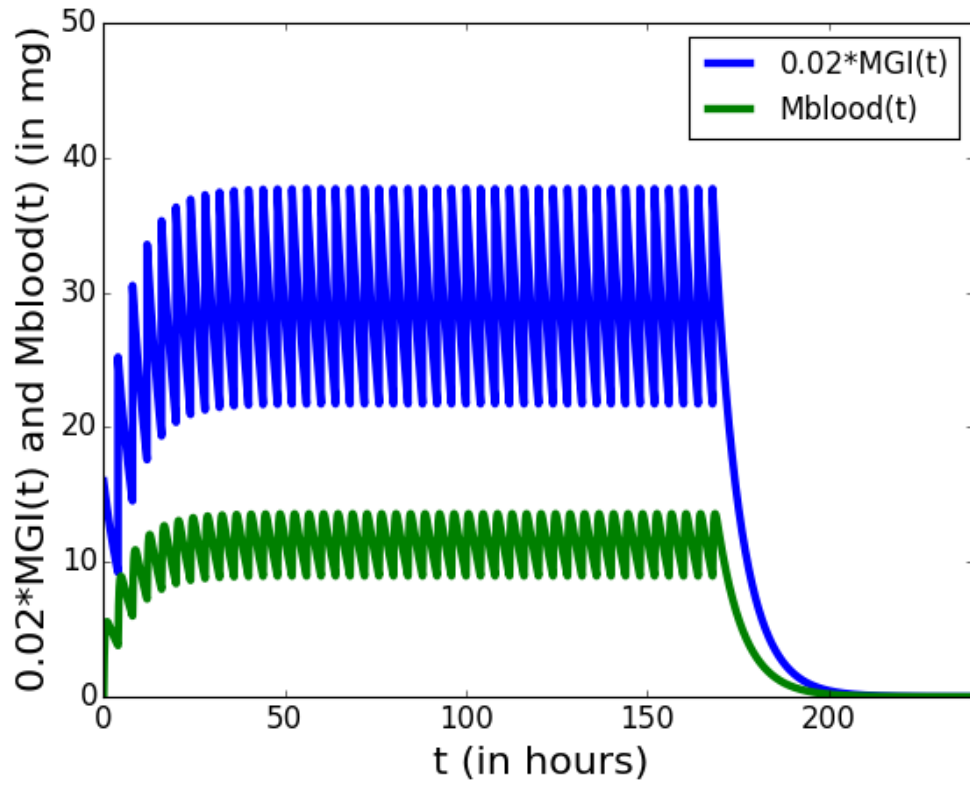


Figure 4.4: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 4.1 and 4.2 Complete usual 7 day treatment and the three of following days;  $M_{GI}(t)$  was scaled in a 1:50 ratio

## Chapter 5

# Inclusion of a variable volume of the gastrointestinal tract contents

In order to accurately model the intake of aciclovir in non-fasting conditions, the volume of the gastrointestinal tract contents have to vary with time. This chapter deals with the inclusion of such variable volume. In section 5.1 a simple model for the emptying of the gastrointestinal tract is presented. In section 5.2 appropriate values for the constants concerning such model are selected and determined. In 5.3 this model is incorporated to the system of differential equations presented in chapter 4. Finally, in chapter 5.4 the results from a couple of simulations are presented.

### 5.1 A simple model for the emptying of the gastrointestinal tract

After ingestion, the volume of the stomach begins to decrease in a linear fashion, as suggested by the data recovered by Mackie R. A. [8] Generalizing this the results for the entire gastrointestinal tract, the situation would be modeled by equation 5.1, where  $d_4$  is the slope.

$$\frac{dV_{GI}(t)}{dt} = -d_4 \quad (5.1)$$

According to equation 5.1,  $V_{GI}(T)$  decreases indefinitely. As more time passes the volume should tend to a value;  $V_{GI}(T)$  should be bounded. Equation 5.2 is capable of modeling this situation; at first the behavior is quite linear, as time passes  $V_{GI}(T)$  starts tending more and more to a fixed value  $V_{small}$ ; the

smallest possible volume for the gastrointestinal tract contents.

$$\frac{dV_{GI}(t)}{dt} = -d_4 \left( \frac{V_{GI}(t)}{V_{small}} - 1 \right) \quad (5.2)$$

## 5.2 Selection and determination of appropriate values for the constants

The smallest possible volume for the gastrointestinal tract contents ( $V_{small}$ ) would be the fasting volume used so far; 0.163 L. This value was recovered from a study performed by Schiller C., *et al.*. To achieve this mean fasting volume, the participants of the study did not eat for at least eight hours, so after eight hours we would expect the volume of the gastrointestinal tract contents to be quite close to 0.163 L. In this same study, the participants were given an standardized meal; after an hour the mean volume for the gastrointestinal tract volumes was 0.751 L. [2] If a person were given a little less food than that given in the study; those 0.751 L could be the initial volume after the meal ( $V_{GI}(0) = 0.751L$ ). If that was the case, a  $d_4$  of 0.09 would yield realistic results as the volume of the gastrointestinal tract after eight hours is within a 1% of 0.163 L. Equation 5.2 then yields the results shown in figure 5.1.

## 5.3 Incorporation of a third differential equation to the system

Incorporation equation 5.2 to the lasts chapter system results as follows.

$$\frac{dM_{GI}(t)}{dt} = -d_1 \left( \frac{M_{GI}(t)}{V_{GI}(t)} - \frac{M_{blood}(t)}{V_{blood}} \right) \quad (5.3)$$

$$\frac{dM_{blood}(t)}{dt} = d_1 d_3 \left( \frac{M_{GI}(t)}{V_{GI}(t)} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}} \quad (5.4)$$

$$\frac{dV_{GI}(t)}{dt} = -d_4 \left( \frac{V_{GI}(t)}{V_{small}} - 1 \right) \quad (5.5)$$

## 5.4 Simulations

Figure 5.2 shows the simulation of a single dose taken with a meal. Figure 5.3 shows the simulation of the treatment discussed in past chapters and a usual mealing pattern; that is, a breakfast followed six hours latter by a lunch, a lunch followed six hours latter by a dinner, a dinner followed twelve hours

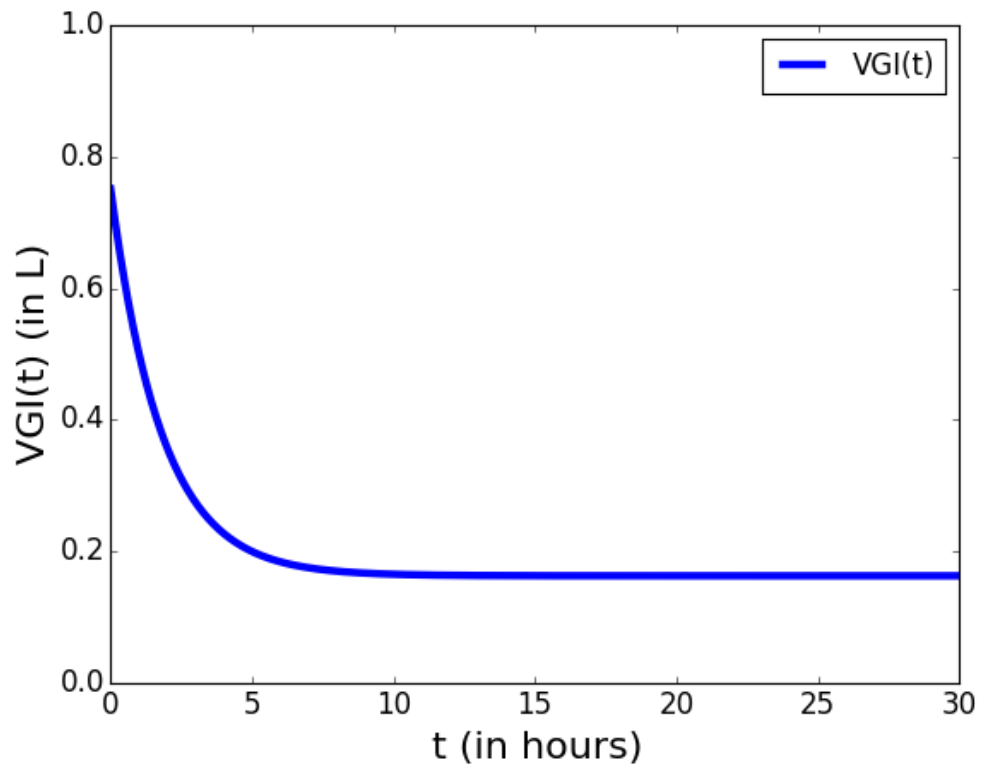


Figure 5.1: **Variation of the gastrointestinal tract volumes with respect to time according to eq. 5.2**  $d_4$  is equal to 0.09 and  $V_{small}$  to 0.163 L



latter by a breakfast and so on. The initial conditions for both simulations were  $M_{GI}(0) = 800mg$ ,  $M_{blood}(0) = 0$  and  $V_{GI}(0) = 0.751L$ . For both figures  $M_{GI}(t)$  was scaled in a 1:50 ratio,  $M_{blood}(t)$  in a 3:2 ration and  $V_{GI}(t)$  in a 4:1 ratio to facilitate visualization.

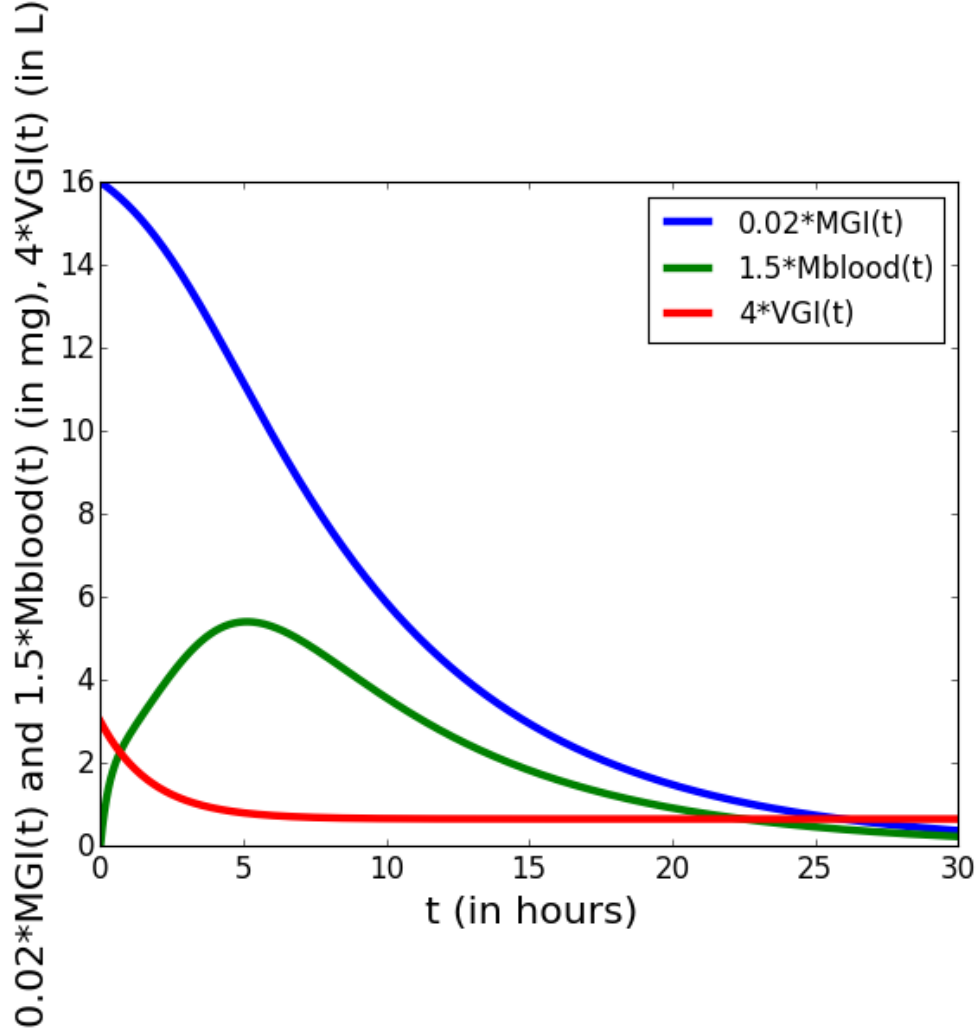


Figure 5.2: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract, as well as the variation of the volume of the gastrointestinal tract with respect to time according to eq. 5.3, 5.4 and 5.5 Single dosage;  $M_{GI}(t)$  was scaled in a 1:50 ratio,  $M_{blood}(t)$  in a 3:2 ration and  $V_{GI}(t)$  in a 4:1

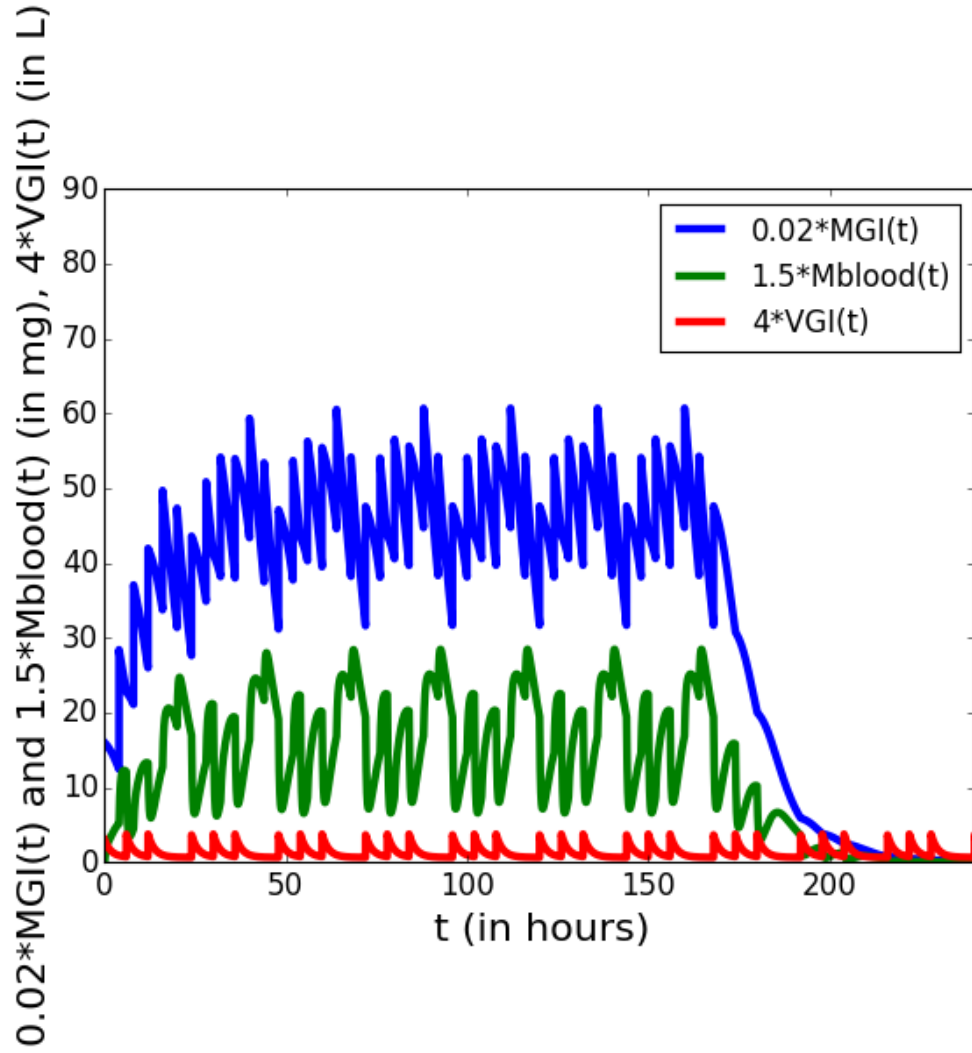


Figure 5.3: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract, as well as the variation of the volume of the gastrointestinal tract with respect to time according to eq. 5.3, 5.4 and 5.5 Complete usual 7 day treatment and the three of following days;  $M_{GI}(t)$  was scaled in a 1:50 ratio,  $M_{blood}(t)$  in a 3:2 ration and  $V_{GI}(t)$  in a 4:1

## Chapter 6

# Improving the predictive capacity of the model

As mentioned in section 4.4 the model for a single take reaches the peak in about two hours instead of one, most likely because the aciclovir pill takes time to dissolve. Section 6.1 describes a way to account for pill dissolution for a single dose.

### 6.1 A model that accounts for pill dissolution for the intake of a single pill

When compared to figure 4.2, the data recovered by P. Susantakumar *et al.* seems to be scaled horizontally by a factor of two before the peak; after the peak, it seems not to be scaled at all. To perform this scaling, the right hand side of equations 4.1 and 4.2, as they represent slopes, would have to be multiplied by 0.5. However this should happen just for times prior to two hours; else it should be multiplied by 1. So the situation would be better modeled by equations 6.1 and 6.2.

$$\frac{dM_{GI}(t)}{dt} = -kd_1 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \quad (6.1)$$

$$\frac{dM_{blood}(t)}{dt} = kd_1d_3 \left( \frac{M_{GI}(t)}{V_{GI}} - \frac{kM_{blood}(t)}{V_{blood}} \right) - kd_2 \frac{M_{blood}(t)}{V_{blood}} \quad (6.2)$$

Where  $k$  is an scaling factor.  $k = 0.5$  if  $t < 2$  and  $k = 1$  if  $t \geq 2$   
To account for non-fasting conditions every  $V_{GI}$  in the above system should be substituted by  $V_{GI}(t)$  and equation 5.5 should be added.

## 6.2 Simulation

The simulation for this system is shown in figure 6.1;  $M_{GI}(t)$  was scaled in a 1:100 ratio. As always, the initial conditions were set to be  $M_{GI}(0) = 800mg$ ,  $M_{blood}(0) = 0$ .

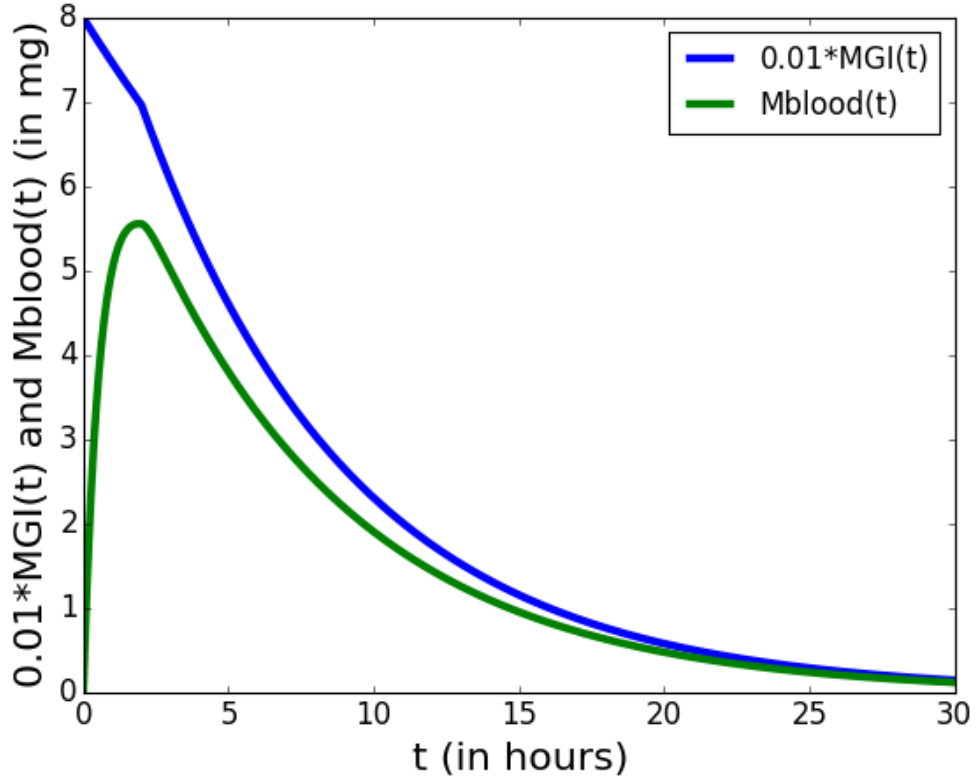


Figure 6.1: Variation of aciclovir mass in blood and in the contents of the gastrointestinal tract with respect to time according to eq. 6.1 and 6.2 Single dosage;  $M_{GI}(t)$  was scaled in a 1:100 ratio

## Chapter 7

# Conclusions

The model proposed in chapter two started simple. From there, in each new cycle, more elements were incorporated. Starting from the model in the third cycle (chapter 4) valuable predictions could be made; in this case for fasting conditions. The model in the Fourth cycle (chapter 5) is valuable for non-fasting conditions. In the last modeling cycle (chapter 6) a more complicated but more precise model was developed to yield results that closely resemble the real dynamics of aciclovir throughout the gastrointestinal tract and the blood.

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# Appendix A

## Python code

The following Python code can be used to generate any of the results obtained in this report. In order to fit, long lines of code were divided into several in. For the code to function, after copying the code from this appendix and then pasting it into a python editor, those lines should have to be put together into a single line again. Specifically, when pasting the code, lines 10 through 12 and 85 and 86 should receive such treatment. Be careful; after editing lines 10 through 12, lines 85 and 86 will become lines 83 and 84. It is also important to do this with the comments as there are some references to line numbers in the code.

It is crucial to notice that, when the code is properly edited as described above, the code will produce accurate results when conserving either lines 42 through 45 or 36 and 37 but not both. This is because the model described in chapter 6 just works for the intake of a single pill.

Additional guidance is included in the code's comments.

```
# -*- coding: utf-8 -*-
# Program      : Euler's method for a system
# Author       : MOOC team Mathematical Modelling Basics
# Created      : May, 2017
# Modified by  : Rodrigo Gonzalez Linares

import numpy as np
import matplotlib.pyplot as plt

print("Solution for dMGI/dt=d1*((MGI(t)/VGI(t))-(Mblood(t)/Vblood)),
dMblood/dt=d1*d3*((MGI(t)/VGI(t))-(Mblood(t)/Vblood))-d2*(Mblood(t)/Vblood),
dVGI(t)/dt=-d4*((VGI(t)/Vsmall)-1)")

# Initializations
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Dt = 0.01 # timestep Delta t
MG_init = 800 # initial MG (mass of aciclovir in the gastrointestinal tract
contents)
MB_init = 0 # initial Mb (mass of aciclovir in the blood)
VG_init = 0.751 # initial VG (volume of the gastrointestinal tract contents);
#to simulate fasting or constant MG/VG set to 0.163
t_init = 0 # initial time
t_end = 250 # stopping time

n_steps = int(round((t_end-t_init)/Dt)) # total number of timesteps

X = np.zeros(3) # create space for current X=[MG,MB,VG]^T
dXdt = np.zeros(3) # create space for current derivative
t_arr = np.zeros(n_steps + 1) # create a storage array for t
X_arr = np.zeros((3,n_steps+1)) # create a storage array for X=[MG,MB,VG]^T
t_arr[0] = t_init # add the initial t to the storage array
X_arr[0,0] = MG_init # add the initial MG to the storage array
X_arr[1,0] = MB_init # add the initial MB to the storage array
X_arr[2,0] = VG_init # add the initial VG to the storage array

# Euler's method
c=1
for i in range (1, n_steps + 1):
    t = t_arr[i-1] # load the time
    if t>1 and round(t,5)\%4==0 and round(t,5)<169: #Adds one pill to the
#system every 4 hours till the 10th day. Remove to symulate a single pill take.
#NOTE: it is not comatible with the scaling of MG and MB (REMOVE LINES 42
#TROUGH 45)
        X_arr[0,i-1]=X_arr[0,i-1]+800
    if t>1 and round(t,5)\%6==0: #Adds volume to the gastrointestinal tract
#contents (simulating mealing) at times 0,6,12,24,30,36,... Remove to symulate
#fasting conditions
        c=c+1
        if c\%4!=0:
            X_arr[2,i-1]=X_arr[2,i-1]+.751
    if t<2: #Scales MG and MB vs. t are scaled horizontally in a 2:1 ratio till
#times lower than 2 and in a 1:1 in times higher than 2. NOTE: it is not
#compatible for several pill takes (REMOVE LINES 36 AND 37)
        c=0.5
    else:
        c=1
    MG = X_arr[0,i-1] # load the value of MG
    MB = X_arr[1,i-1] # load the value of MB
    VG = X_arr[2,i-1] # load the value of VG

```



```

X[0] = MG                                # fill current state vector X=[MG,MB,VG]^T
X[1] = MB
X[2] = VG
dMGdt = -.0225*c*((MG/VG)-(MB/5)) # calculate the derivative dMG/dt; to
#simulate constant MG/VG remove MG and substitute it for 800
dMBdt = .2*.0225*c*((MG/VG)-(MB/5))-17.4*c*(MB/5) # calculate the derivative
#dMB/dt; to simulate constant MG/VG remove MG and substitute it for 800
dVGdt = -.09*(((VG)/0.163)-1) # calculate the derivative dVG/dt
dXdt[0] = dMGdt # fill derivative vector dX/dt
dXdt[1] = dMBdt
dXdt[2] = dVGdt
Xnew = X + Dt*dXdt                      # calculate X on next time step
X_arr[:,i] = Xnew                       # store Xnew
t_arr[i] = t + Dt                      # store new t-value

# Plot the results

fig = plt.figure()
plt.plot(t_arr, X_arr[0,:], linewidth = 4, label="MGI(t)") # plot MG vs. time;
#to scale MG on the plot multiply X_arr[0,:] by a scaling factor
plt.plot(t_arr, X_arr[1,:], linewidth = 4, label="Mblood(t)") # plot MB vs.
#time; to scale MB on the plot multiply X_arr[1,:] by a scaling factor
plt.plot(t_arr, X_arr[2,:], linewidth = 4, label="VGI(t)") # plot VG vs. time;
#to scale VG on the plot multiply X_arr[2,:] by a scaling factor

plt.title('dMGI/dt=d1*((MGI(t)/VGI(t))-(Mblood(t)/Vblood)),
dMblood/dt=d1*d3*((MGI(t)/VGI(t))-(Mblood(t)/Vblood))-d2*(Mblood(t)/Vblood),
dVGI(t)/dt=-d4*((VGI(t)/Vsmall)-1)', fontsize = 20) # set title
plt.xlabel('t (in hours)', fontsize = 20) # name of horizontal axis
plt.ylabel('MGI(t) and Mblood(t) (in mg), VGI(t) (in L)', fontsize = 20) # name
#of vertical axis

plt.xticks(fontsize = 15)                # adjust the fontsize
plt.yticks(fontsize = 15)                # adjust the fontsize
plt.axis([0, 30, 0, 16])                 # set the range of the axes

plt.legend(fontsize=15)                  # show the legend
plt.show()                              # necessary for some platforms

# save the figure as .png
fig.savefig('project_aciclovir.png', dpi=fig.dpi, bbox_inches = "tight")

```